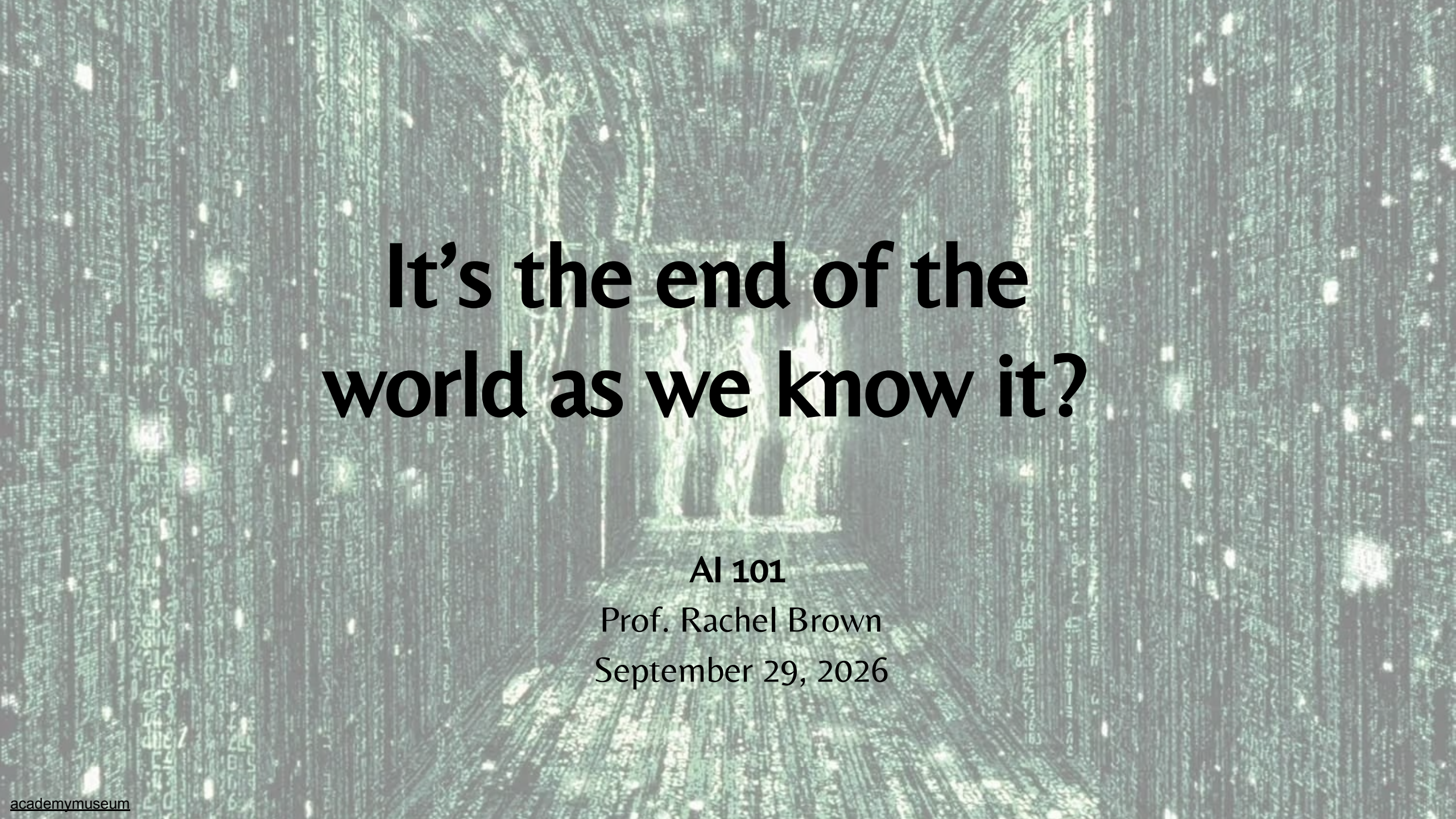




It's the end of the world as we know it?

AI 101

Prof. Rachel Brown

September 29, 2026

About Me



Dr. Rachel Brown

Assistant Professor of Computer Science

School of Computing and Information Science

Willamette University

Previously at NVIDIA and UC Berkeley

Research Interests

Human-interpretable AI

Alternatives to pixel representations of images

rabrown3@willamette.edu

Today's Agenda

What is a neural network?

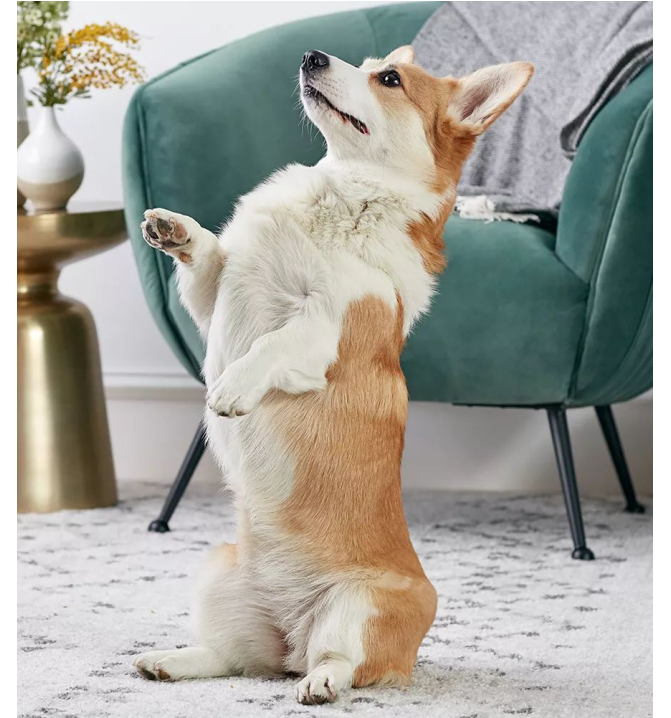
Why are neural networks suddenly so powerful?

Is AI going to cause the end of the world next year?

What does a neural network do?

Input data  Calculate an answer

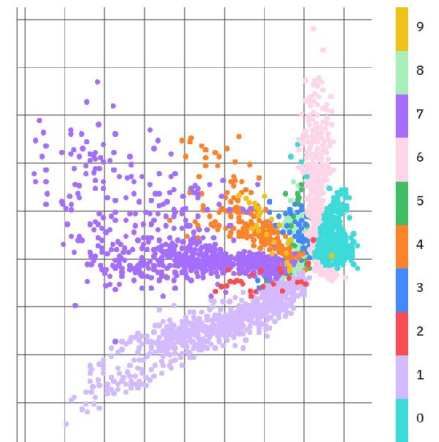
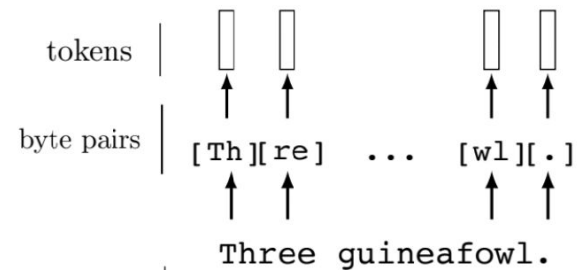
To do this successfully, the network must first be **trained** with ~~treats~~ criticism & compliments



What is a Neural Network?

After training a network, you will have 3 things:

1. A prediction machine
2. An embedding
3. A latent space





The Prediction Machine

Input data



Input Vector

X1	11
X2	12
X3	13
X4	14
X5	15
X6	16
X7	17
X8	18

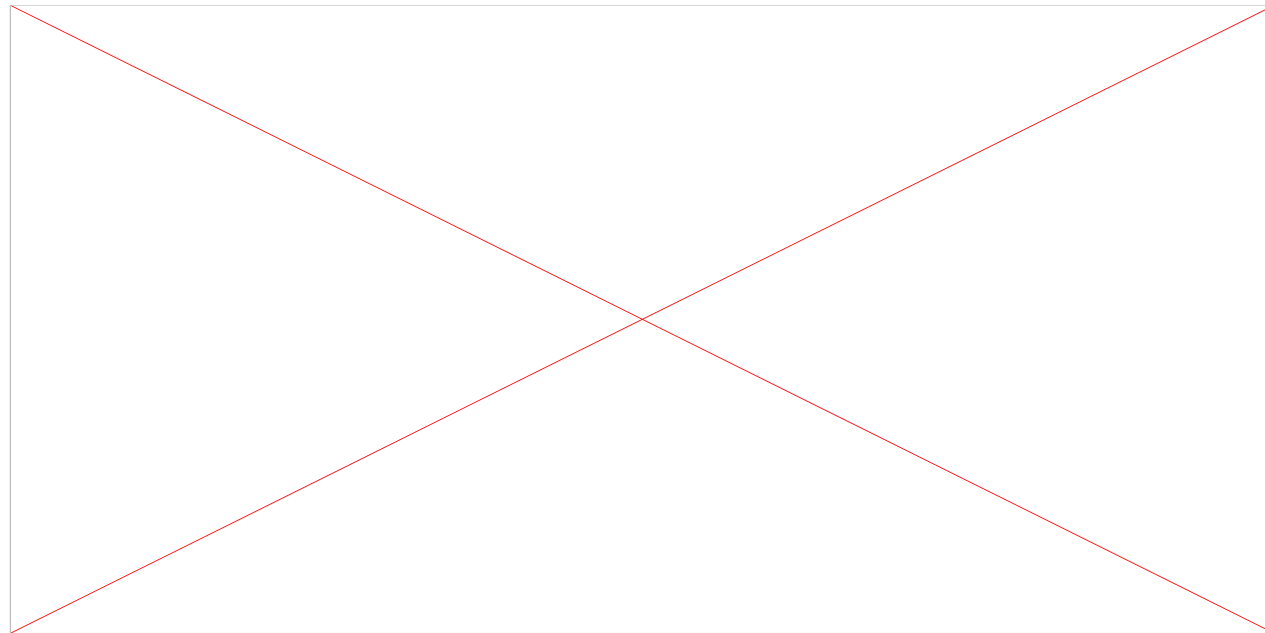
Do y'all know what a vector is?

A set of numbers describing the coordinates of a point in space

2D vector $[x, y]$

3D vector $[x, y, z]$

nD vector $[x, y, z, a, b, c, \dots]$





The Prediction Machine

Input data

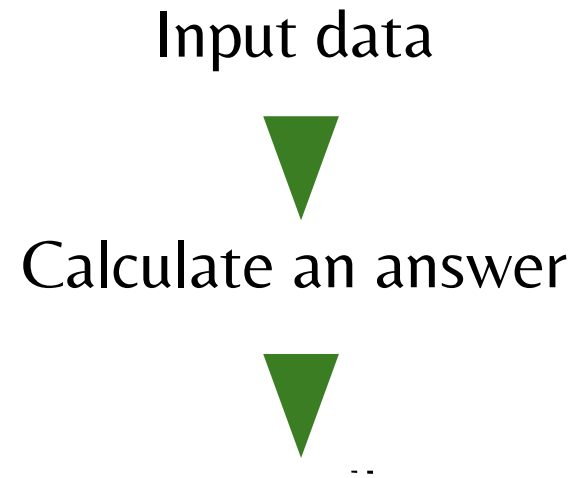


Input Vector

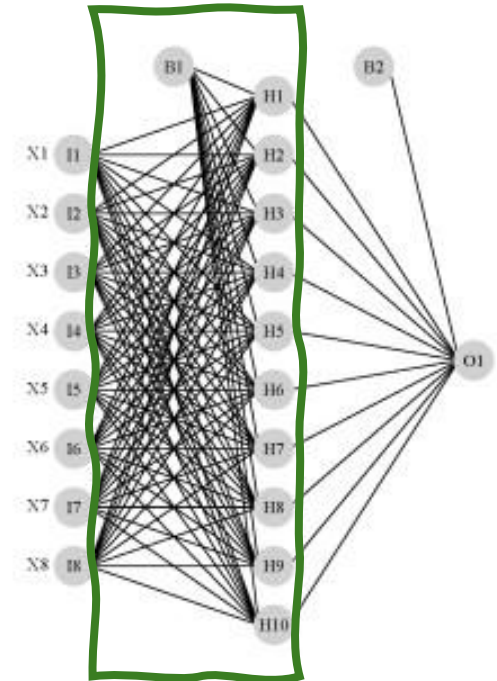
X1	11
X2	12
X3	13
X4	14
X5	15
X6	16
X7	17
X8	18



The Prediction Machine

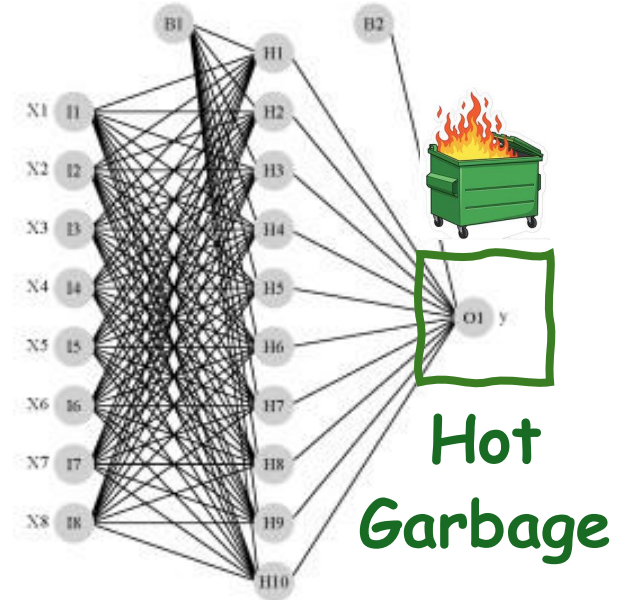
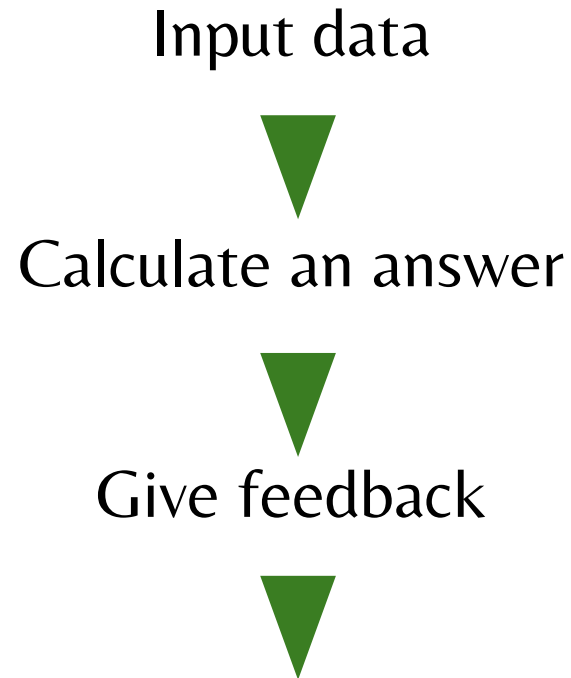


Random



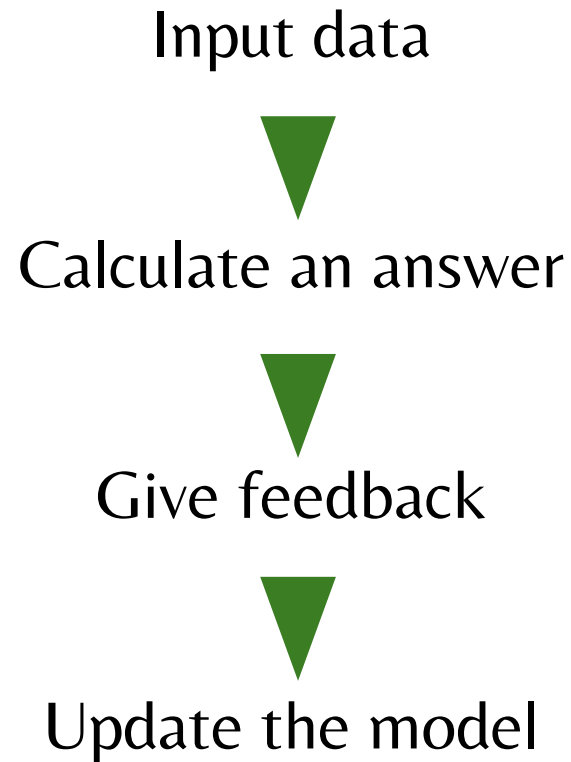


The Prediction Machine

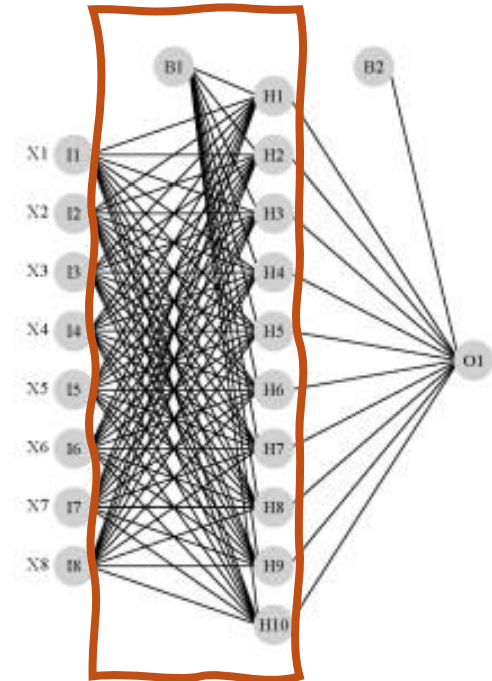




The Prediction Machine



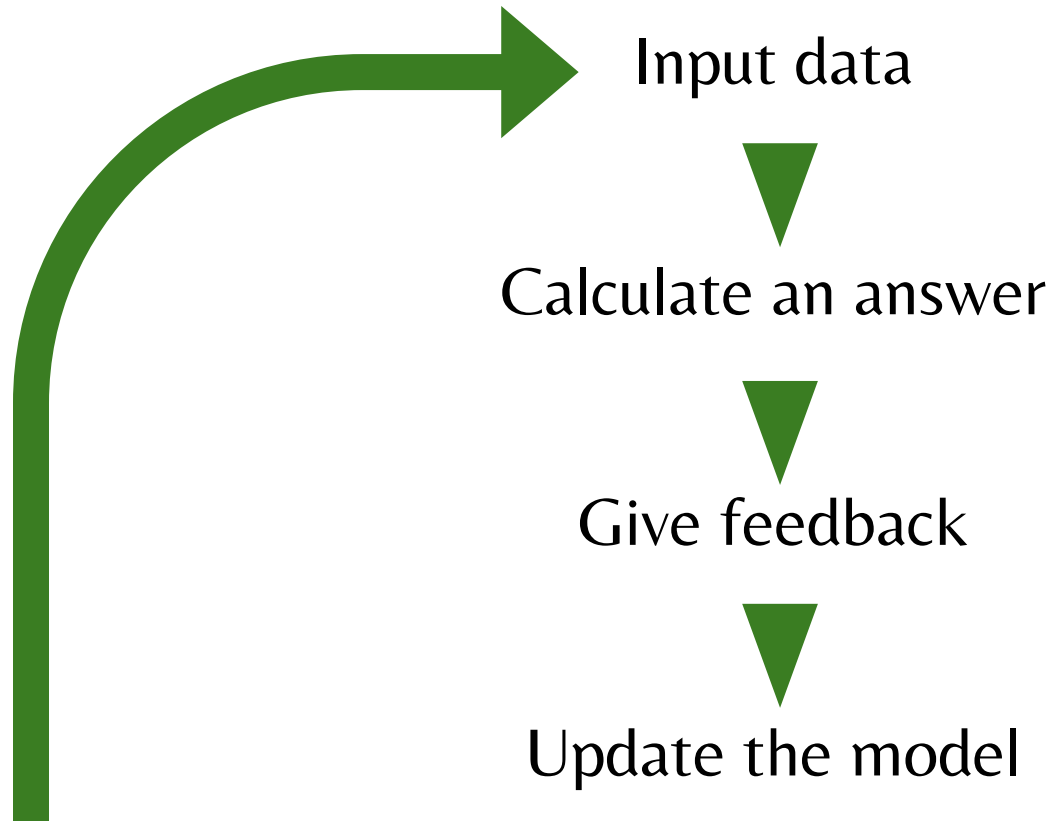
Updated!



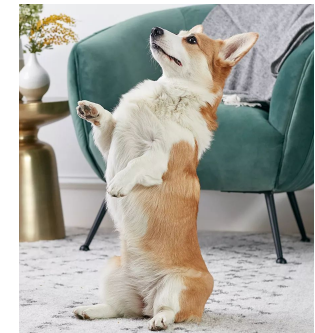
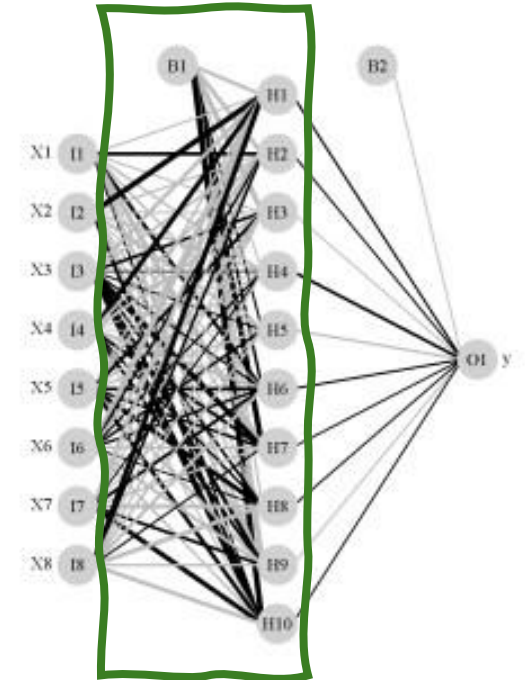
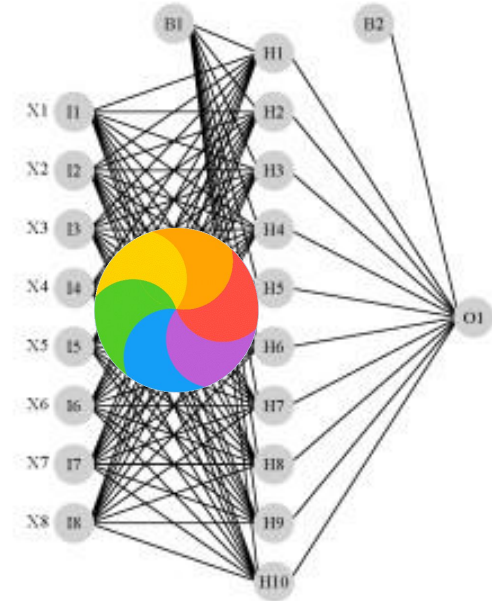


The Prediction Machine

Trained Weights

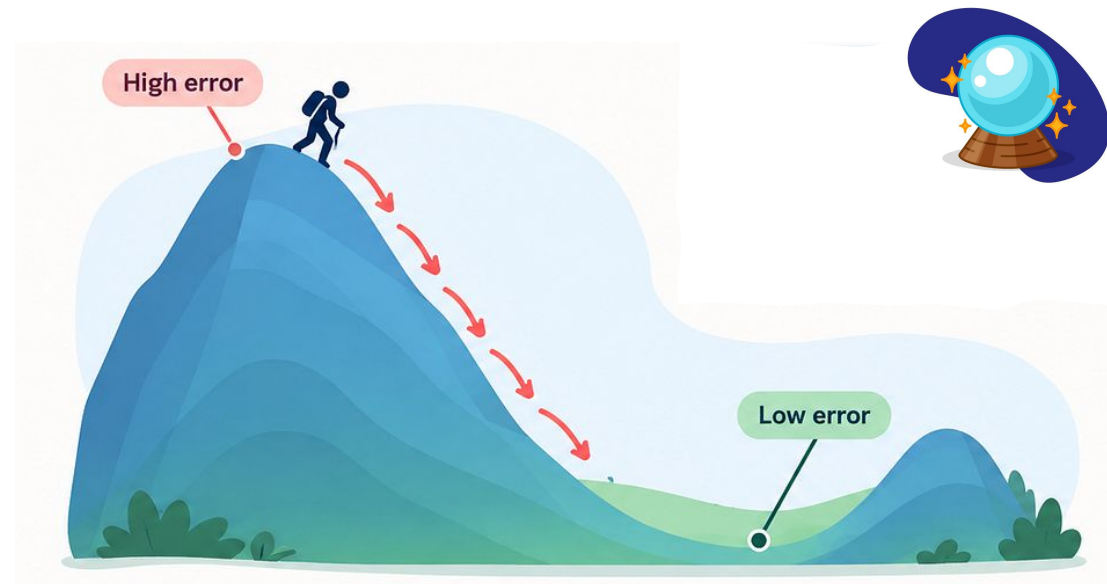


Repeat until model behavior meets minimum standards

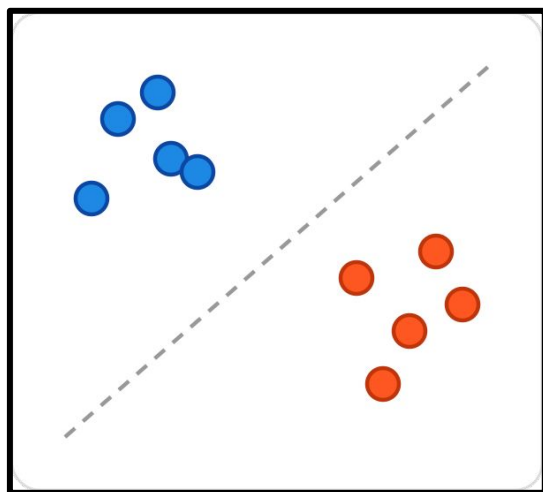


The Prediction Machine

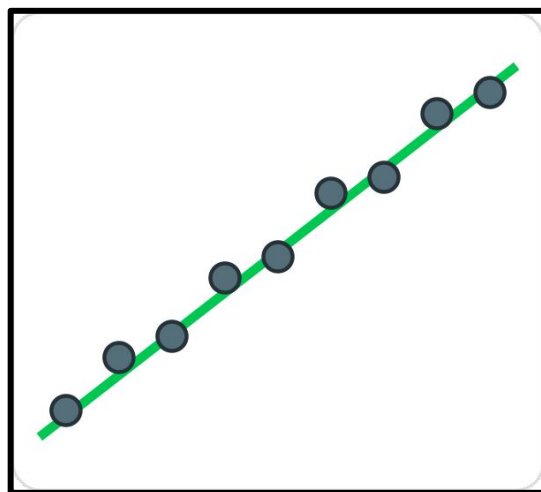
All neural networks follow this same basic training loop, called **gradient descent**



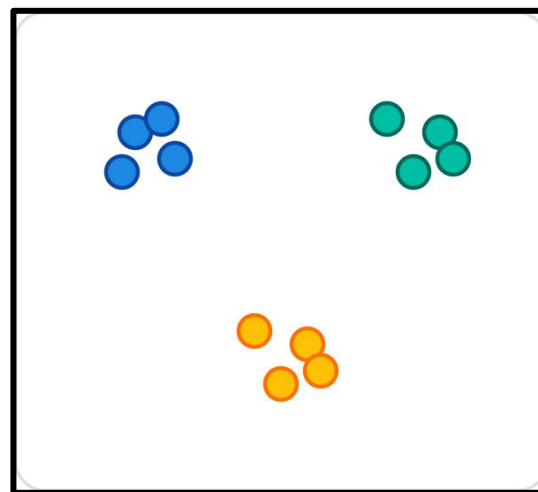
Classification



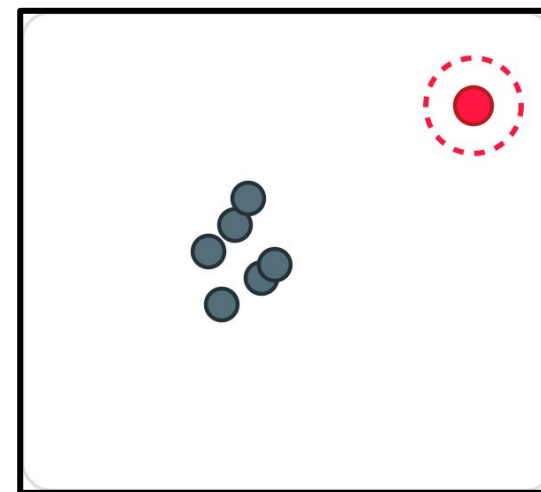
Regression



Clustering



Anomaly Detection



What is a Neural Network?

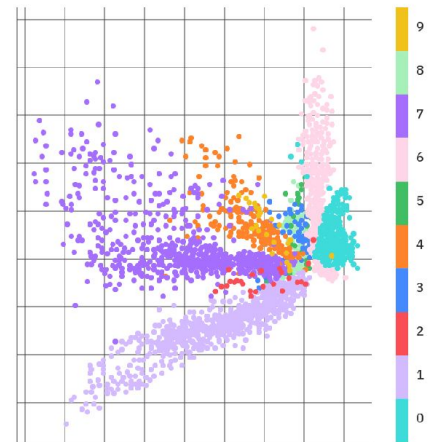
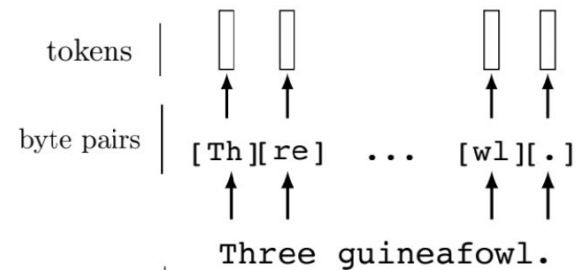
After training, you will have 3 things:

1. A prediction machine



2. An embedding

3. A latent space

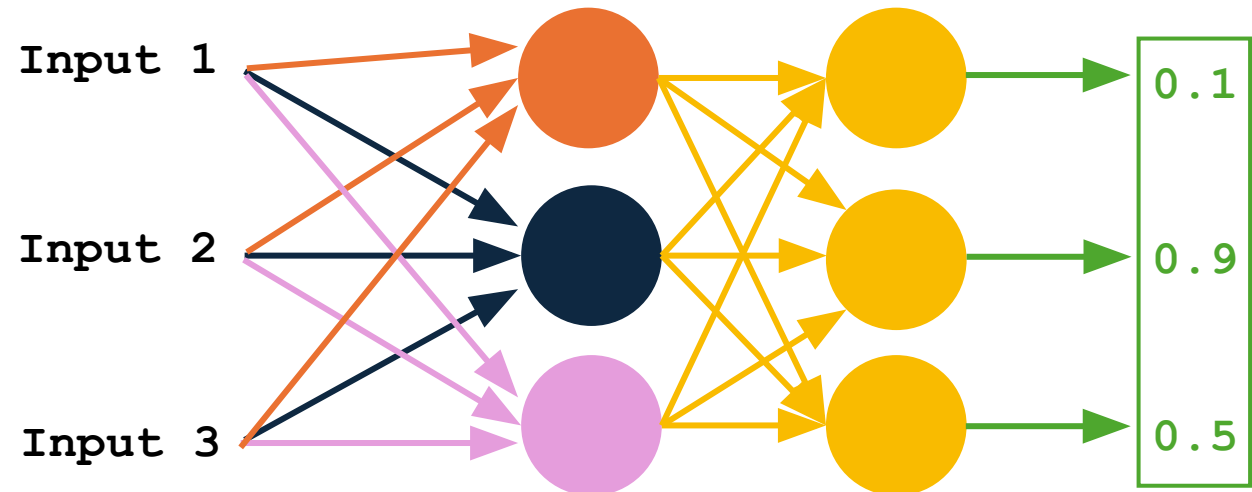
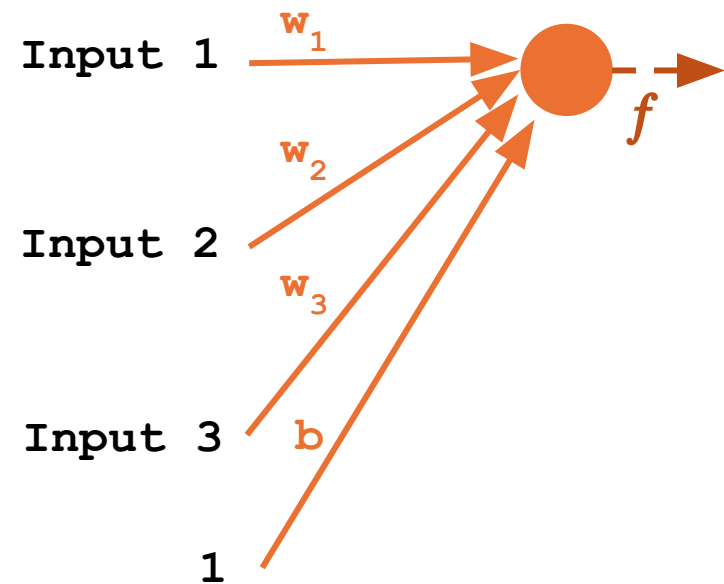


Neural Network Model

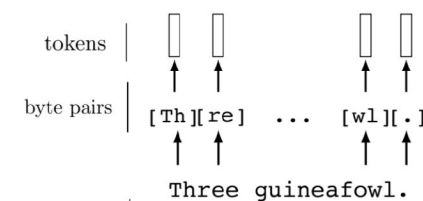
Connections between neurons are described as **weights**

Each neuron also has a **bias** term (for efficiency) and a non-linear **activation function** (for greater complexity)

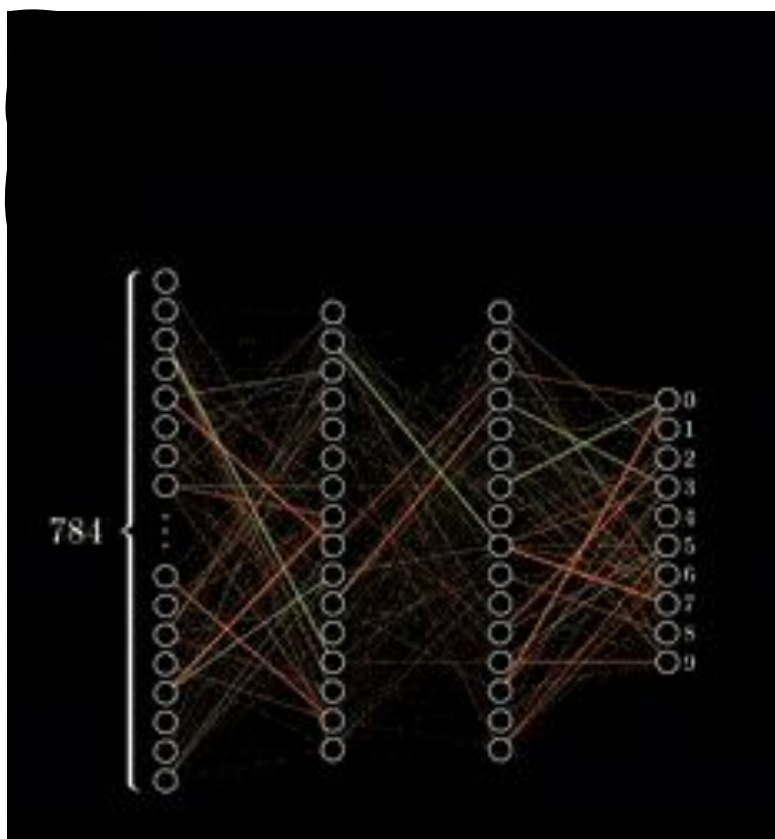
Multiple sets of neurons connected in series are called **layers**

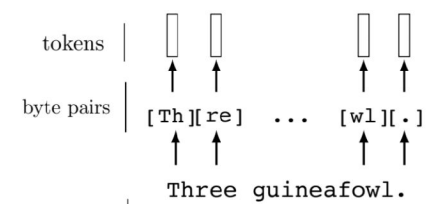


The Embedding



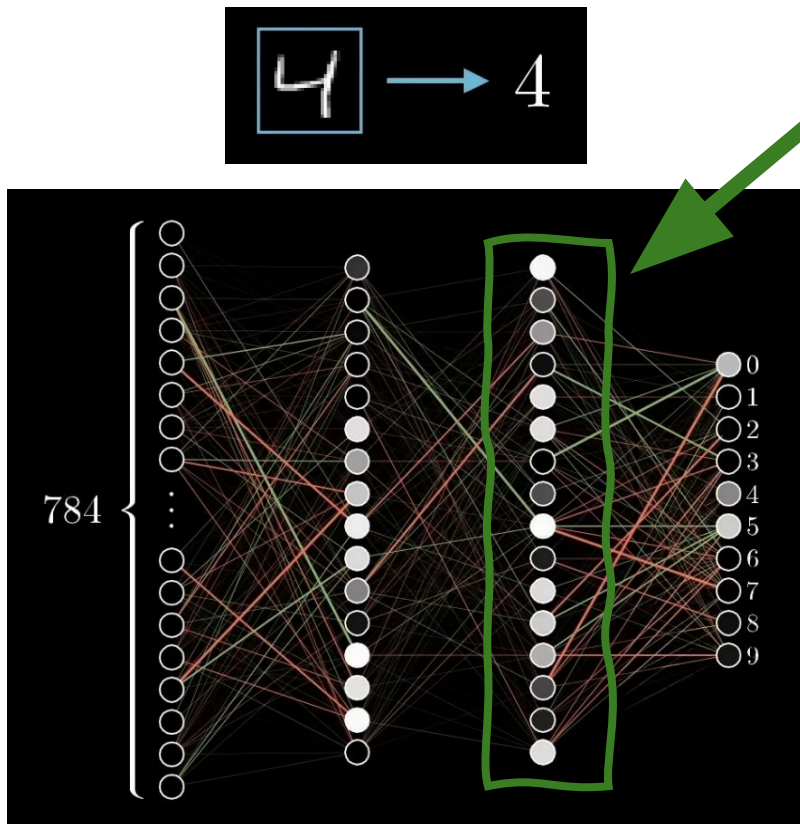
Each input vector produces a different activation pattern in the network





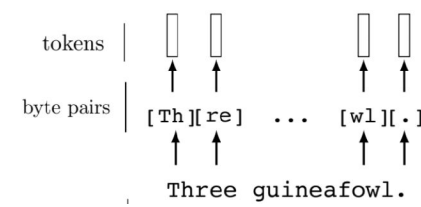
The Embedding

Each input vector produces a different activation pattern in the network



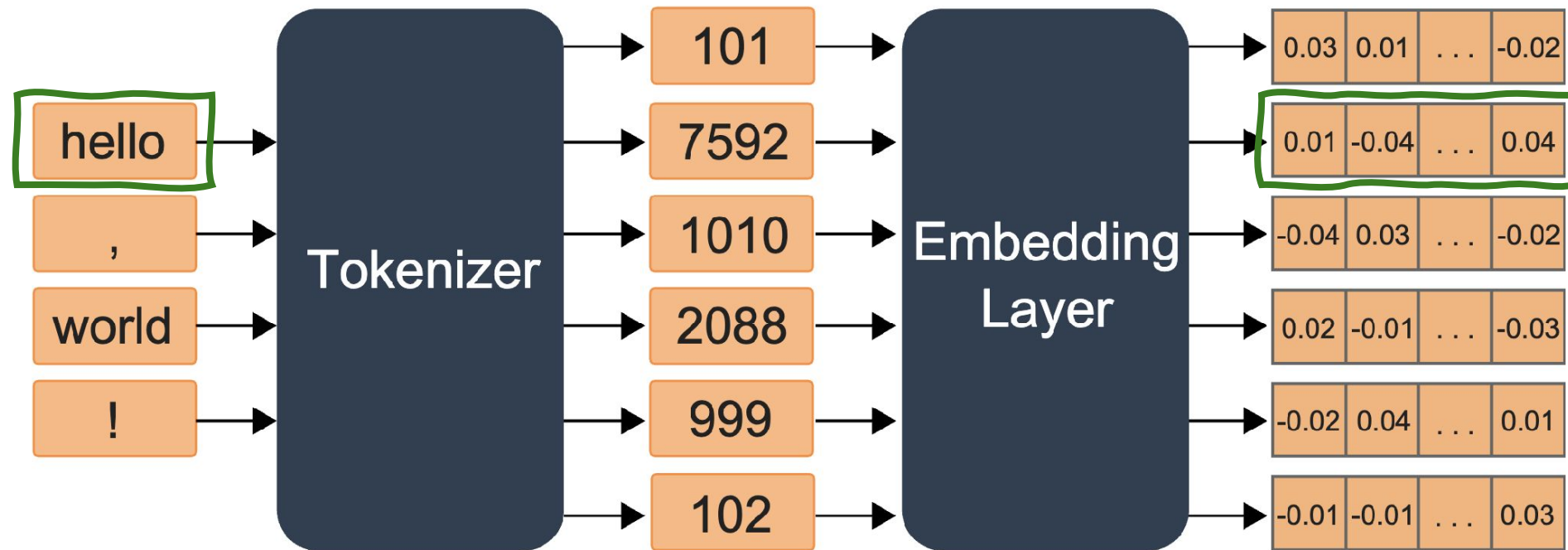
The network activation pattern for a given input is called the **embedding** of that input

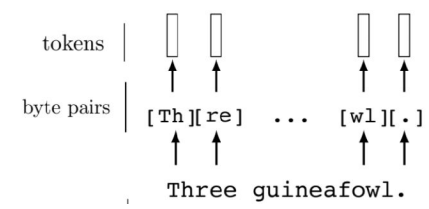
We usually use the activation pattern of the 2nd-to-last layer in the network



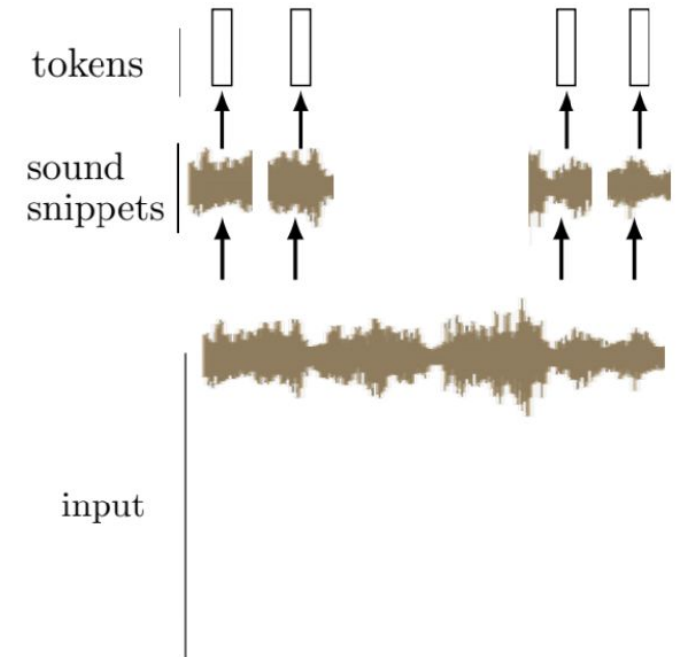
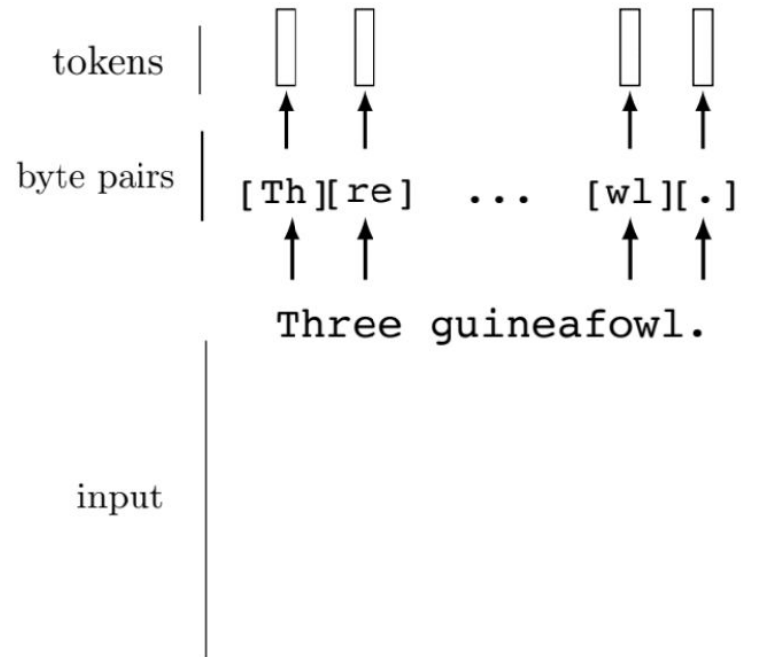
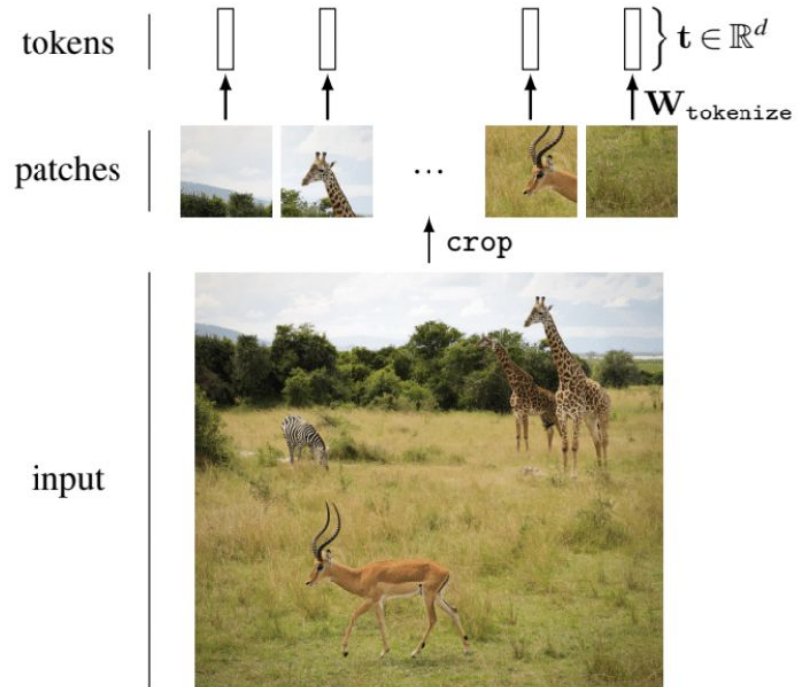
The Embedding

Embeddings are always vectors (or matrices) of continuous numeric values





You can create an embedding from anything!



What is a Neural Network?

After training, you will have 3 things:

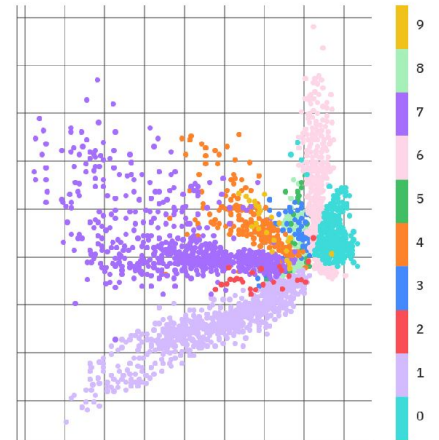
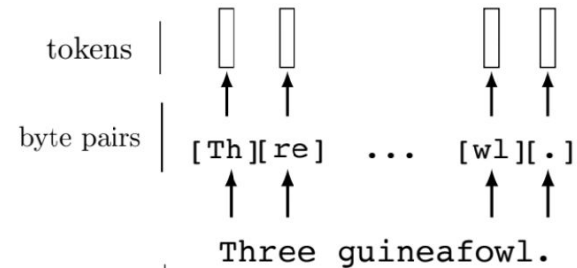
1. A prediction machine

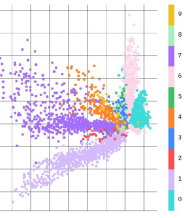


2. An embedding



3. A latent space

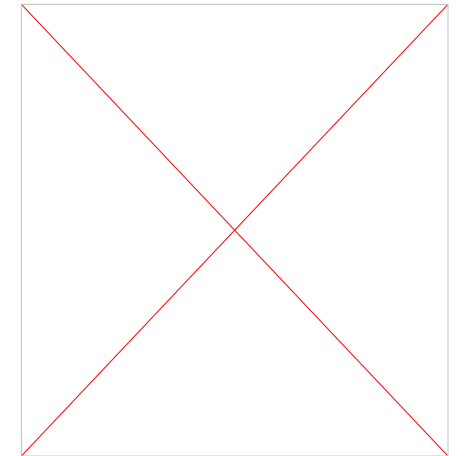
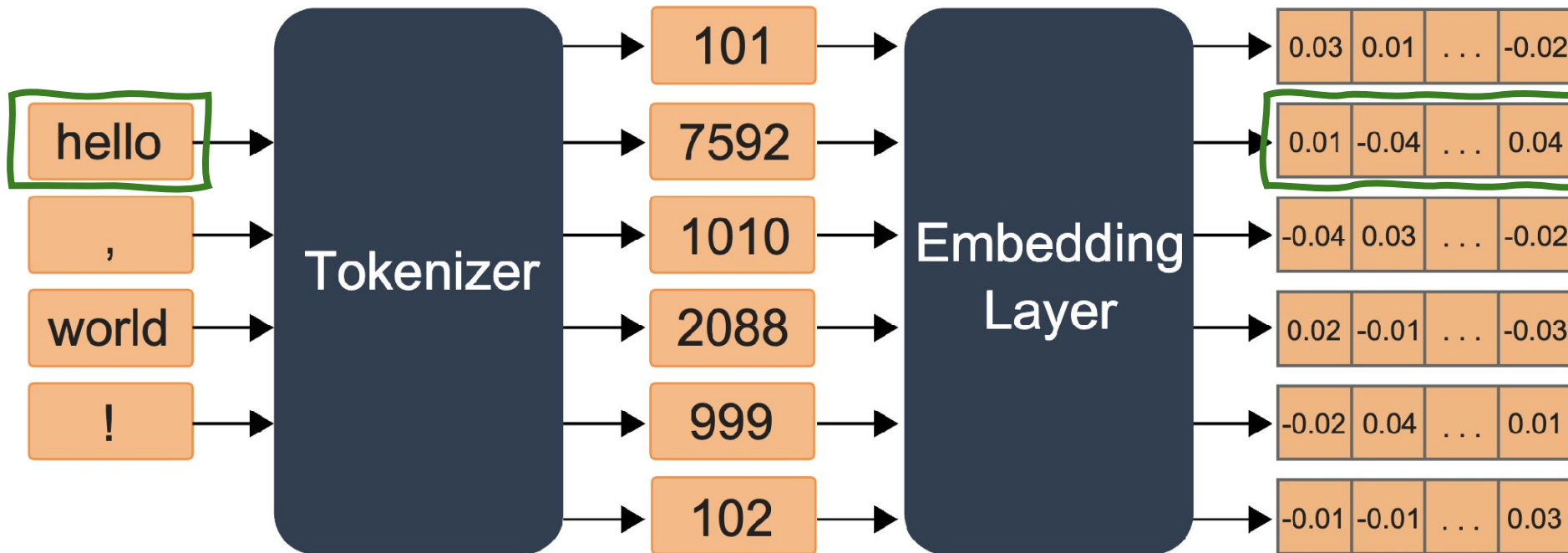




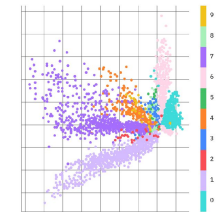
The Latent Space

Embeddings are always vectors (or matrices) of continuous numeric values

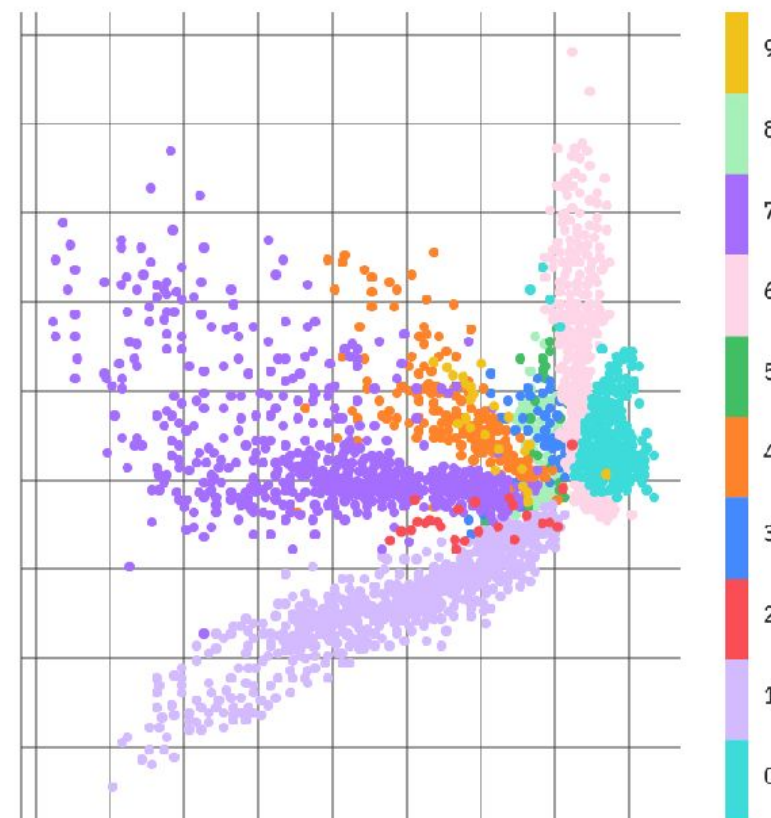
A **vector** of values implies a **location** in space



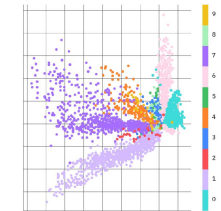
The Latent Space



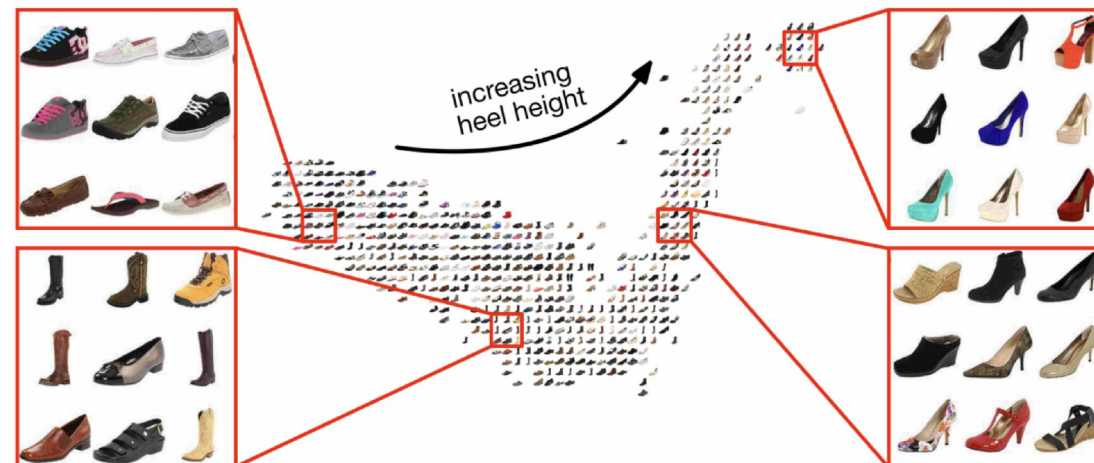
Locations in the latent space correspond to specific types of input



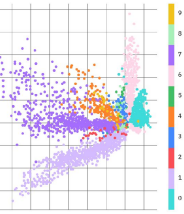
The Latent Space



Distances in the latent space
encode similarity

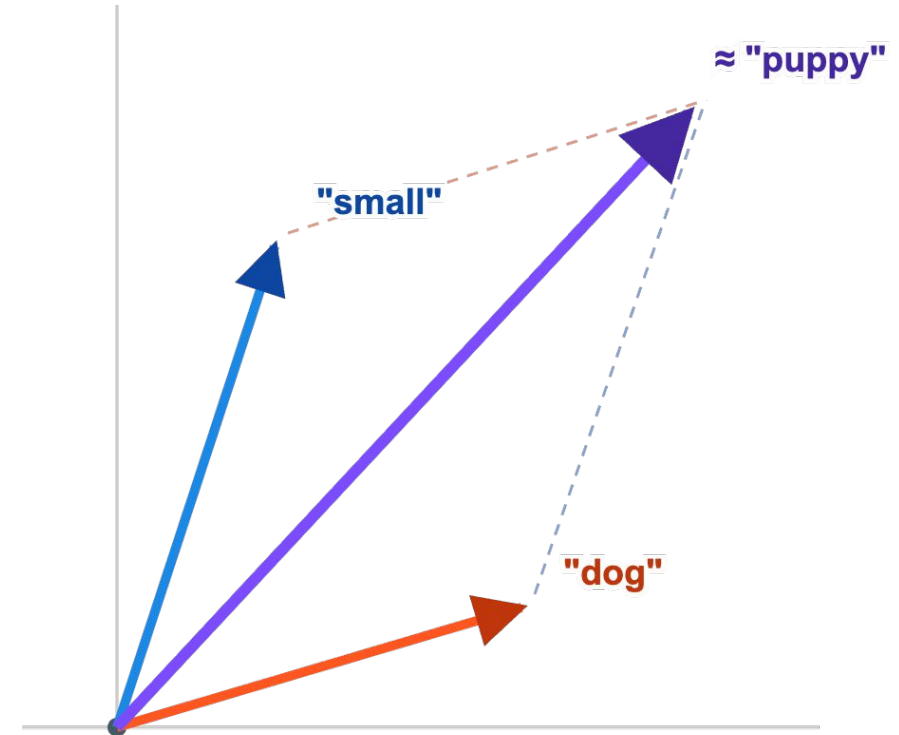


The Latent Space



Directions in the latent space
encode concepts

Two embeddings added together
results in a combination of those
two concepts



What is a Neural Network?

After training, you will have 3 things:

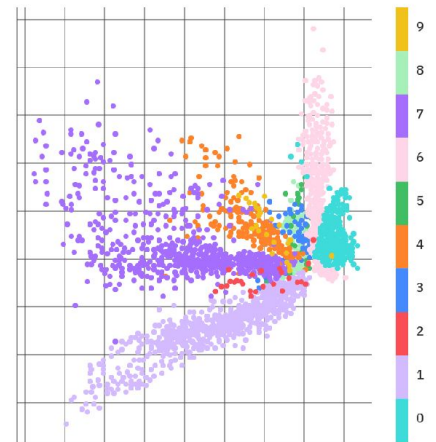
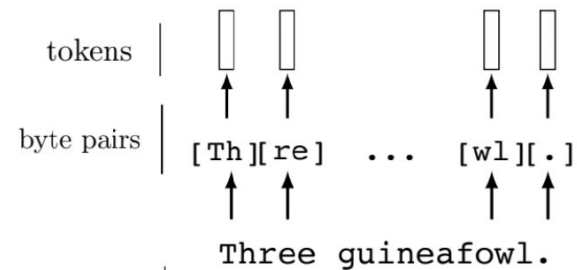
1. A prediction machine



2. An embedding



3. A latent space



Today's Agenda

~~What is a neural network?~~

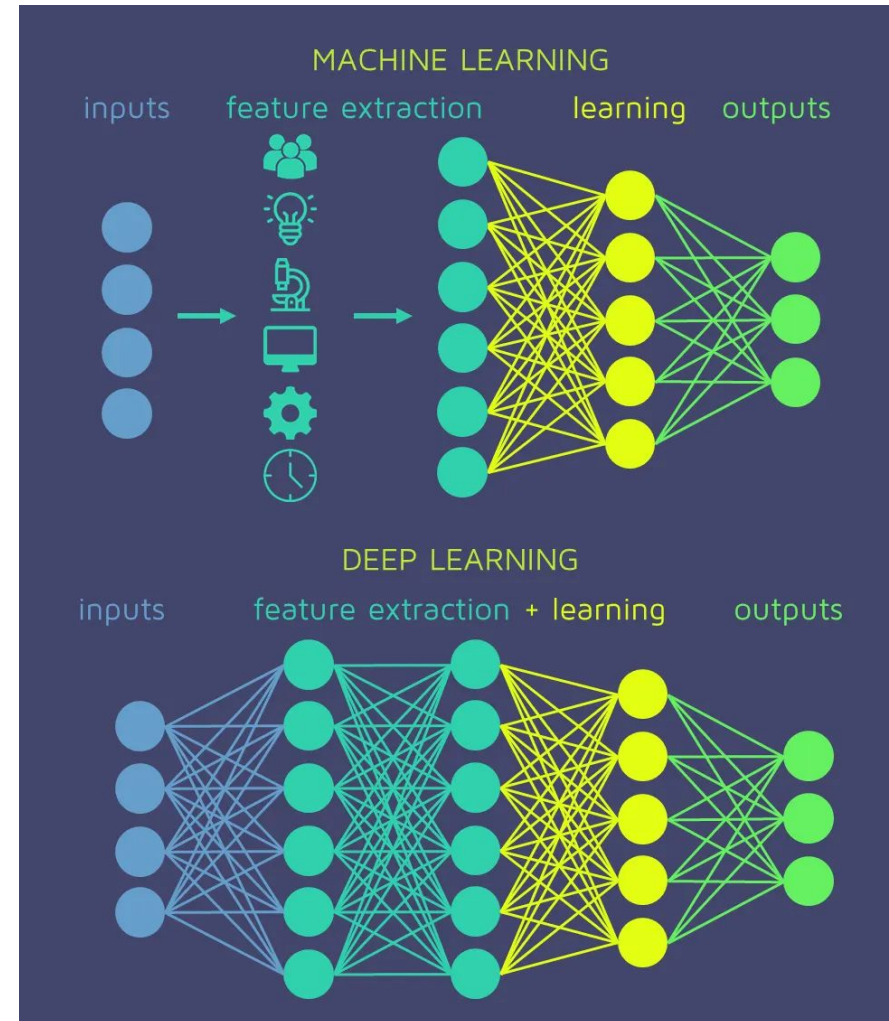
Why are neural networks suddenly so powerful?

Is AI going to cause the end of the world next year?

Describing Data Quantitatively

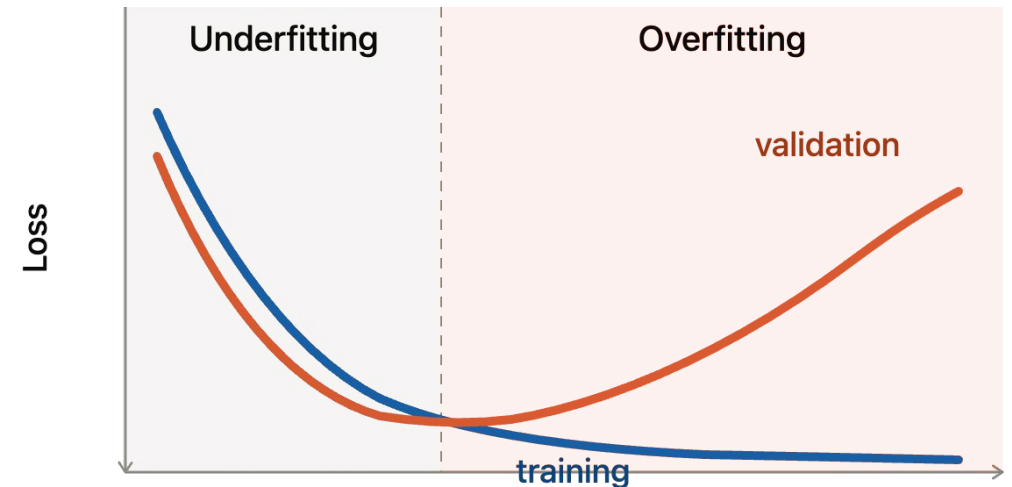
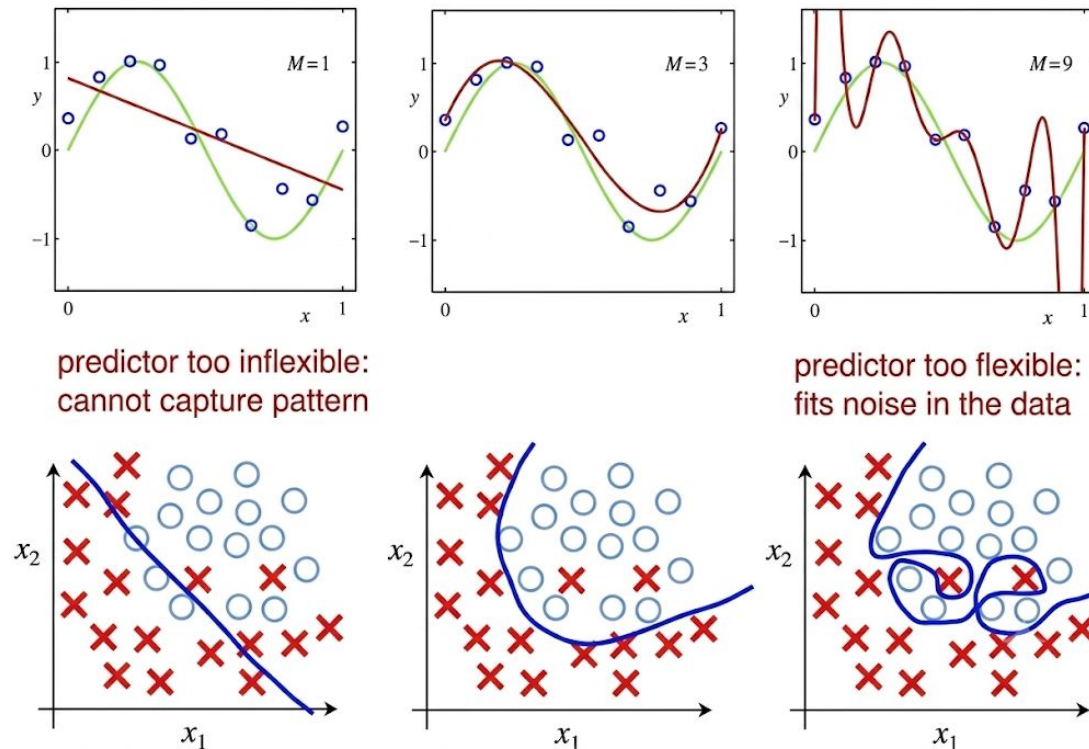
Classical Machine Learning
is about **engineering**
data representations

Modern Deep Learning
is about **learning**
data representations



Learning Requires LOTS of Data

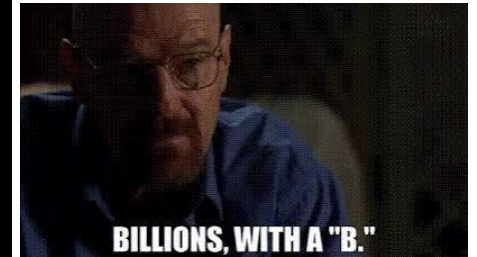
Overly complex models do not generalize well on small datasets



Learning Requires LOTS of Data

Neural networks are VERY complex models with lots of parameters

GPT-3		Total weights: 175,181,291,520
Embedding	$\overset{12,288}{d_embed} * \overset{50,257}{n_vocab}$	= 617,558,016
Key	$\overset{128}{d_query} * \overset{12,288}{d_embed} * \overset{96}{n_heads} * \overset{96}{n_layers}$	= 14,495,514,624
Query	$\overset{128}{d_query} * \overset{12,288}{d_embed} * \overset{96}{n_heads} * \overset{96}{n_layers}$	= 14,495,514,624
Value	$\overset{128}{d_value} * \overset{12,288}{d_embed} * \overset{96}{n_heads} * \overset{96}{n_layers}$	= 14,495,514,624
Output	$\overset{12,288}{d_embed} * \overset{128}{d_value} * \overset{96}{n_heads} * \overset{96}{n_layers}$	= 14,495,514,624
Up-projection	$\overset{49,152}{n_neurons} * \overset{12,288}{d_embed} * \overset{96}{n_layers}$	= 57,982,058,496
Down-projection	$\overset{12,288}{d_embed} * \overset{49,152}{n_neurons} * \overset{96}{n_layers}$	= 57,982,058,496
Unembedding	$\overset{50,257}{n_vocab} * \overset{12,288}{d_embed}$	= 617,558,016

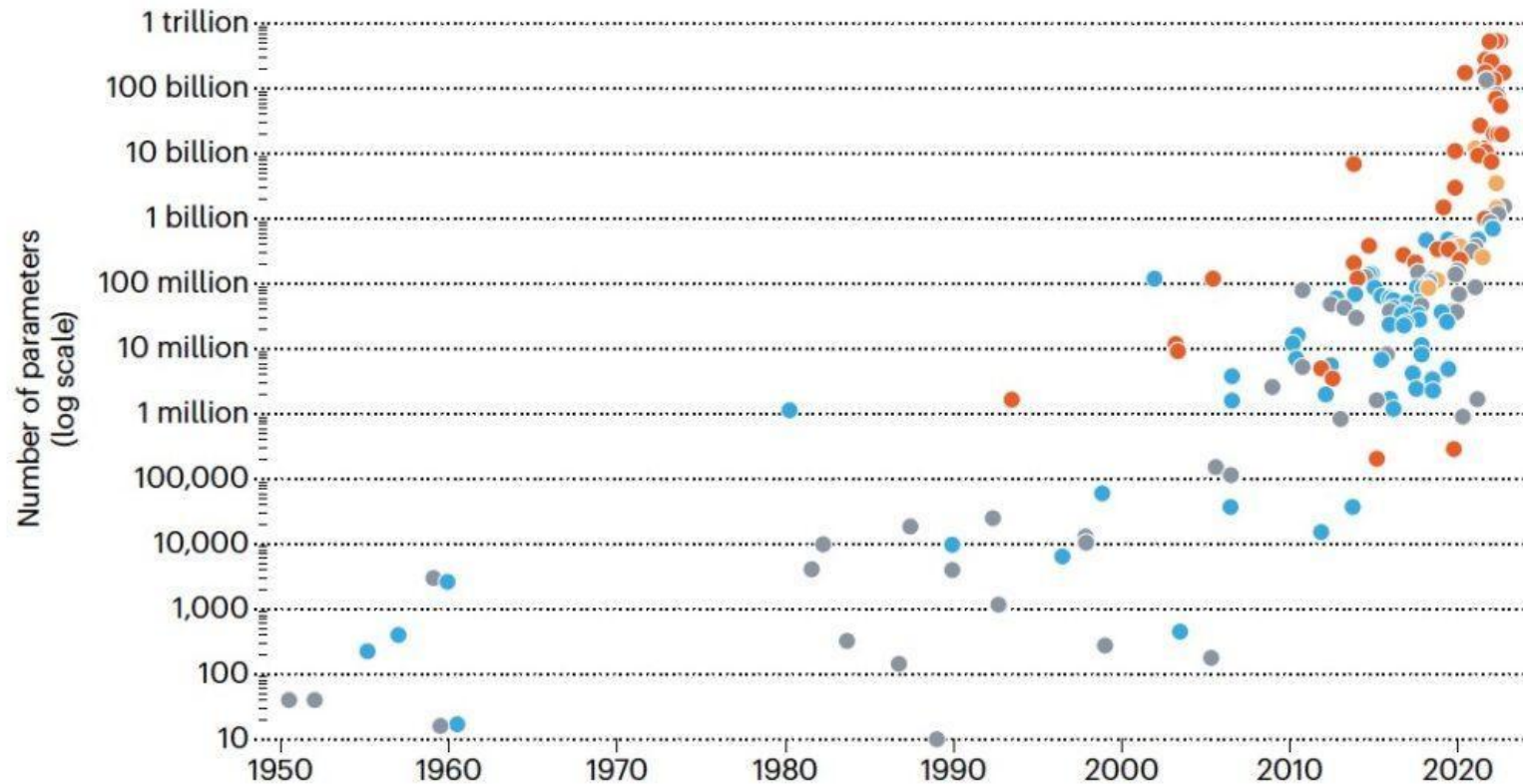


Parallelization Makes Big Data Possible

THE DRIVE TO BIGGER AI MODELS

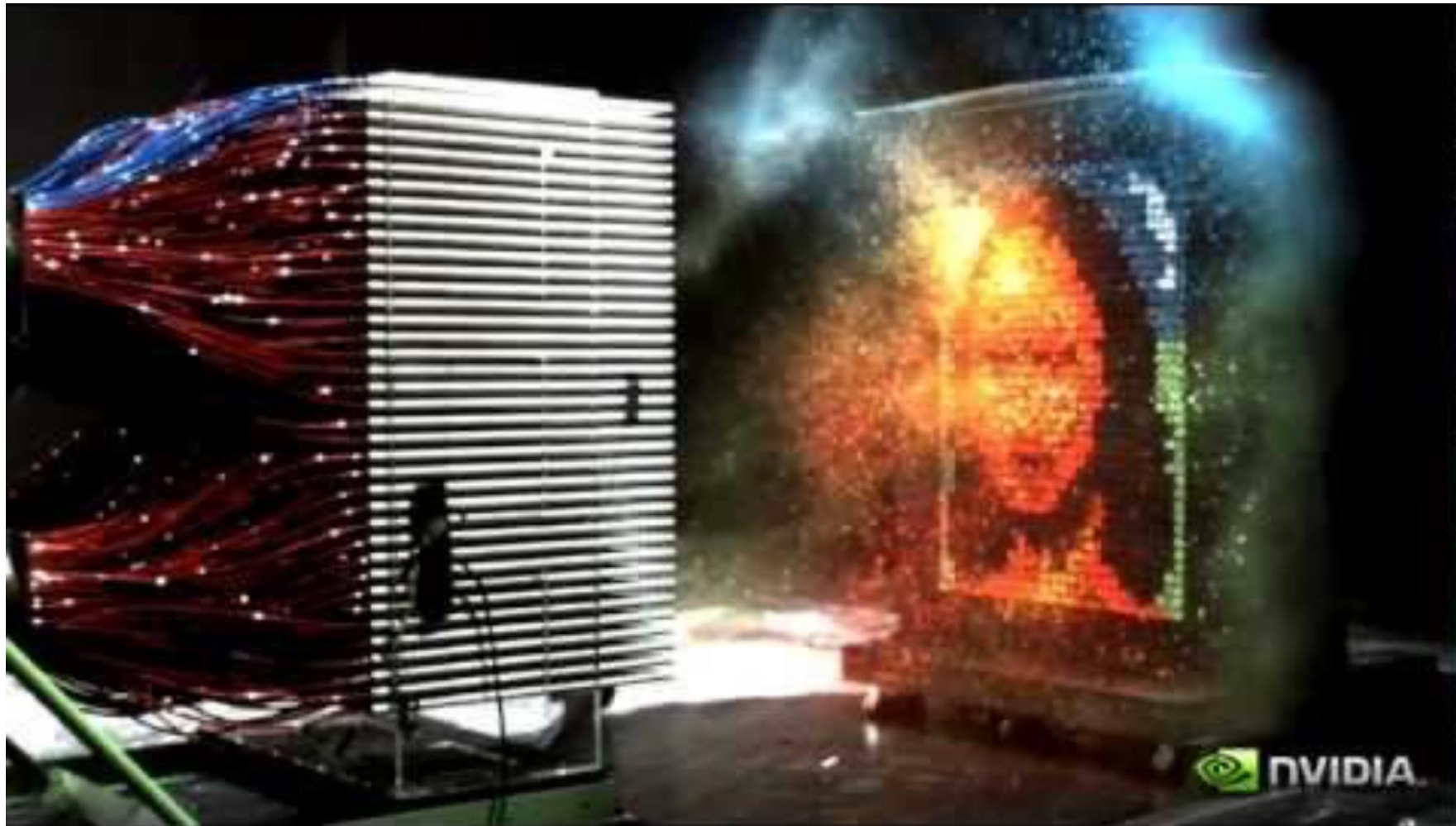
The scale of artificial-intelligence neural networks is growing exponentially, as measured by the models' parameters (roughly, the number of connections between their neurons)*.

● Language ● Image generation ● Vision ● Other



*'Sparse' models, which have more than one trillion parameters but use only a fraction of them in each computation, are not shown.

Parallelization Makes Big Data Possible



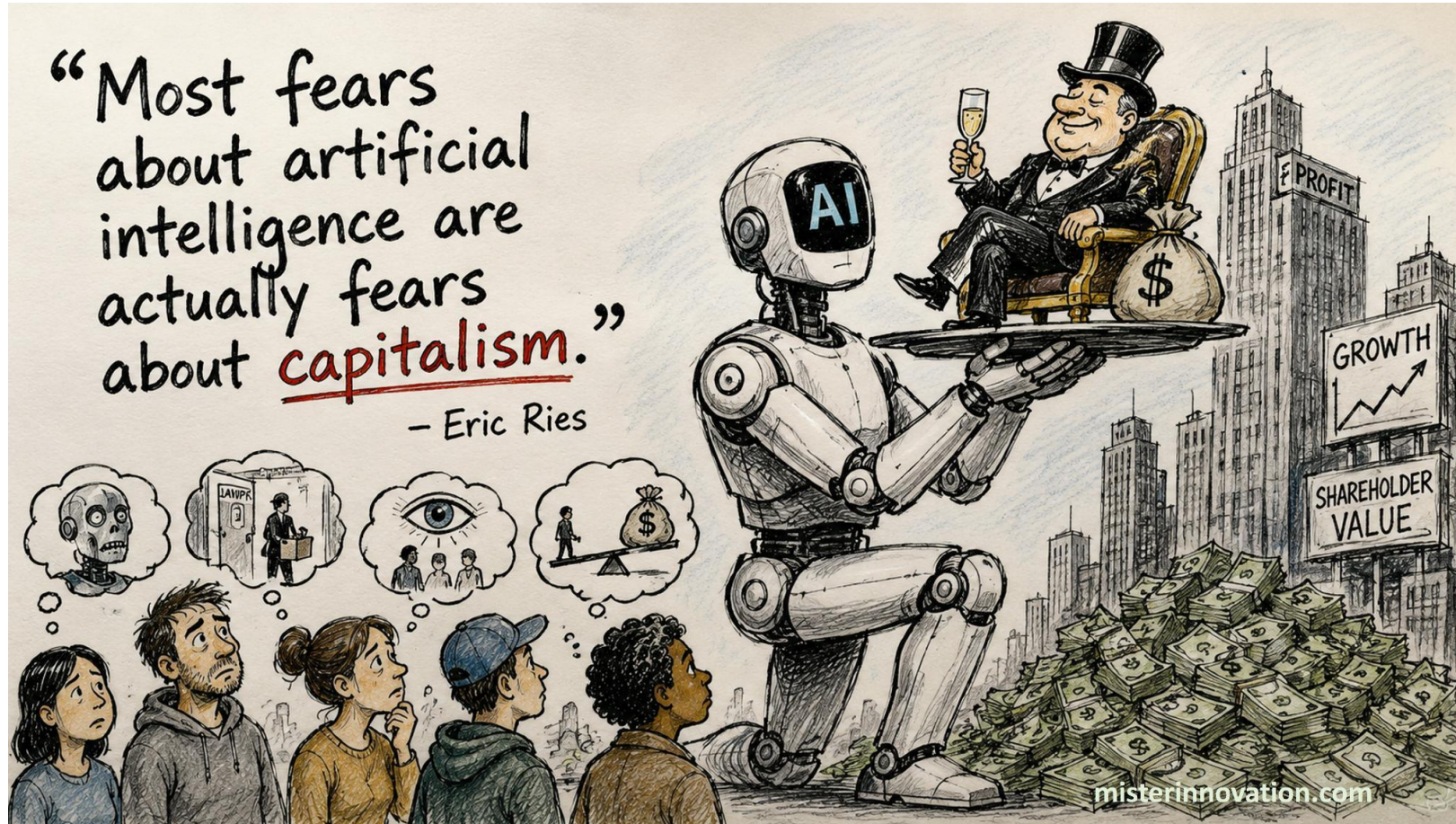
Today's Agenda

~~What is a neural network?~~

~~Why are neural networks suddenly so powerful?~~

Is AI going to cause the end of the world next year?

Is AI going to cause the end of the world?



Is AI going to cause the end of the world?

Probably not.

However, there are two recent technological innovations that seem to be unambiguously bad:

AI ability to mimic humans

Larger AI models that require more data

AI Mimicking Humans

Innovation:

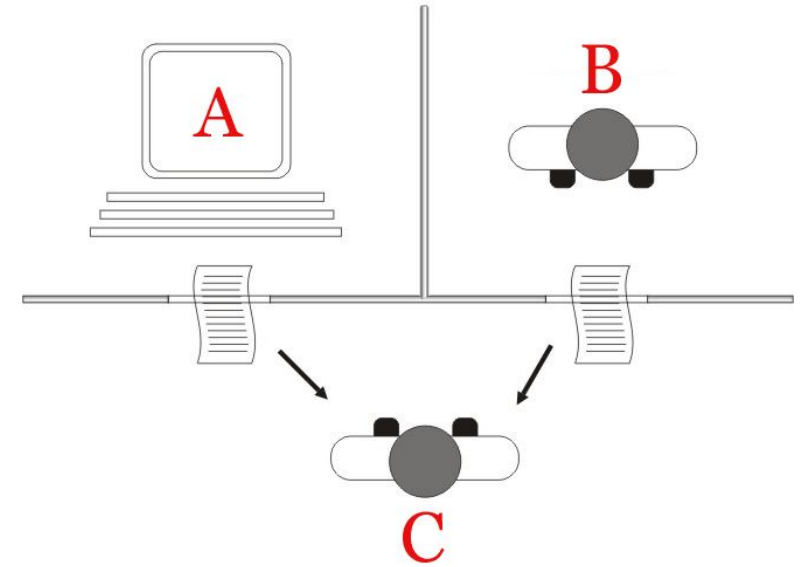
Copying human voice & language patterns

Consequences:

Individual manipulation (shopping, mental health)

Individual scams / misinformation / fake blackmail

Security breaches for infrastructure / military targets



Data Hungry Models

Innovation:

Larger AI models that require more data

Consequences:

Mass surveillance (cameras everywhere)

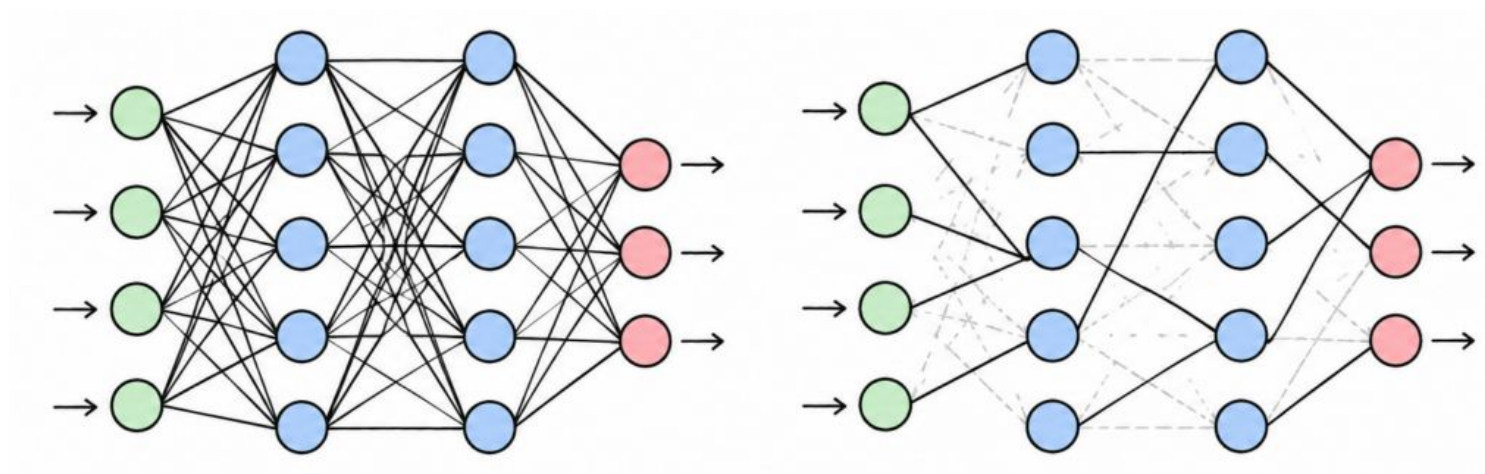
Data mining (identity theft, scams)

Individual manipulation (shopping, voting)

Tech & Govt incentives are aligned!



The Lottery Ticket Hypothesis



Reasons Not to Worry... Yet.

Critical evaluation, creativity, and exploration are NOT solved

Optimizing for results $\neq$ Optimizing for understanding

Many world-ending scenarios require human involvement

Embodied AI (robots) are still very early in development

[An excellent video with citations](#)

Staying Current in AI News & Research

YouTube

- **Fireship (@fireship)**
- **Dwarkesh Patel (@DwarkeshPatel)**
- Julia Turc (@juliaturc1)
- Benn Jordan (@BennJordan)
- 3Blue1Brown (@3blue1brown)
- Jeremy Howard (@howardjeremyp)
- Steve Brunton (@eigensteve)

Podcasts

- **Uncanny Valley (Wired)**
- **Practical AI**
- Latent Space
- Last Week in AI
- Offline (Pod Save America)
- Machine Learning Street Talk (MLST)
- Google DeepMind

Staying Current in AI News & Research

- Be skeptical!
 - Benchmarks are **not** reliable indicators of practical utility
 - CEOs say a **lot** of crazy things
 - Most news reporters don't have **any** technical expertise
- The best sources are people with verified credentials who don't have a financial incentive to promote a specific point of view